

KisanLink — Demonstration Documentation

SIH 2026 • Problem Statement SIH26132

Connect • Trade • Grow

Strengthening Market Linkages & Price Discovery for Farmers

1. Demonstration Objective

Demonstrate the complete farmer-to-buyer marketplace workflow using the working Python/FastAPI + SQLite implementation.

2. Technology Stack

HTML, CSS, JavaScript • Python • FastAPI • Uvicorn • SQLite

3. Recommended Live Demonstration

Step	Action	Show
1	Farmer Login	Open farmer dashboard.
2	Sell Crop	List crop, quantity, unit, price and location.
3	Buyer Login	Open buyer dashboard.
4	Market Search	Search/filter crop, location and price.
5	Price Discovery	Show average/min/max and trend.
6	Smart Matching	Enter crop, quantity, location and maximum price.
7	Matching	Show ranked farmers and match score.
8	Buyer Request	Send request for a crop.
9	Farmer Requests	Show buyer name, mobile, crop and quantity.
10	Accept/Reject	Process the request.
11	Deal	Create and complete the deal.
12	History	Show completed transaction.

4. Smart Matching

The implemented matching engine uses rule-based scoring. It considers crop compatibility, location, price suitability, available quantity and verification, then ranks suitable available listings.

5. Request-to-Deal Flow

Buyer selects a crop → sends request → farmer reviews request → farmer accepts/rejects → accepted request becomes a deal → deal is completed → completed transaction appears in History.

6. Price Discovery / Trends

The application displays crop price information and trend indicators. The KisanLink backend calculates average, minimum, maximum and latest prices from available product listings.

7. Key Demo Talking Points

Direct farmer–buyer connection; better crop discovery; price visibility; rule-based smart matching; structured request-to-deal workflow; transaction history.

8. Scope Statement

Do not claim blockchain, integrated payments, GPS logistics, AI price forecasting or automatic FPO aggregation. The current Smart Matching feature is rule-based.

9. Local Demo

Frontend: <http://127.0.0.1:5501/index.html>

Backend API: <http://127.0.0.1:8081>

10. Final Checklist

Backend running • Frontend running • Clean demo database • Farmer login • Buyer login • Crop listing • Search • Price discovery • Smart matching • Request/Accept/Deal • History